

SAFETY DATA SHEET

SECTION 1: IDENTIFICATION

PRODUCT NAME: MOXY SS WHEEL CLEANER
PRODUCT USE: NON ACID WHEEL CLEANER
MANUFACTURER/SUPPLIER: MOXY PERFORMANCE
ADDRESS: 104 CHALLENGER CT SANFORD, FL 32771
COMPANY TELEPHONE NUMBER: (866) 524-0359
EMERGENCY RESPONSE NUMBER: 1-800-535-5053 (NORTH AMERICA)
+1-352-323-3500 (INTERNATIONAL)

SECTION 2: HAZARD(S) IDENTIFICATION

GHS CLASSIFICATION: HEALTH ENVIRONMENTAL

CLASSIFICATION OF SUBSTANCE OR MIXTURE
PRODUCT DEFINITION: SUBSTANCE

CLASSIFICATION IN ACCORDANCE WITH 29 CFR 1910
(OSHA HCS) AND REGULATION EC NO. 1272/2008
SKIN IRRITATION - CATEGORY 2 [H315]
EYE IRRITATION - CATEGORY 2A [H319]
SPECIFIC TARGET ORGAN TOXICITY, SINGLE
EXPOSURE - CATEGORY 3; STOT SE 3 [H335]

ACUTE TOXICITY, ORAL 0%
ACUTE TOXICITY, DERMAL 0.3%
ACUTE TOXICITY, INHALATION, VAPOR
97.7%
ACUTE TOXICITY, INHALATION, DUST OR
MIST 97.7%

HAZARD STATEMENTS

WARNING!
H315 CAUSES SKIN IRRITATION
H319 CAUSES SERIOUS EYE IRRITATION
H335 MAY CAUSE RESPIRATORY IRRITATION

PICTOGRAM



GHS07

MOXY

PRECAUTIONARY STATEMENTS

GENERAL:

P101 IF MEDICAL ADVICE IS NEEDED, HAVE PRODUCT OR LABEL AT HAND.

P102 KEEP OUT OF REACH OF CHILDREN P103 READ LABEL BEFORE USE.

RESPONSE:

P302+P352 – IF ON SKIN: WASH WITH PLENTY OF SOAP AND WATER

P304+P340+P312 – IF INHALED: REMOVE VICTIM TO FRESH AIR AND KEEP AT REST IN A POSITION COMFORTABLE FOR BREATHING. CALL A POISON CENTER OR DOCTOR IF YOU FEEL UNWELL.

P305+P351+P338 – IF IN EYES: RINSE CAUTIOUSLY FOR SEVERAL MINUTES. REMOVE CONTACT LENSES IF PRESENT AND EASY TO DO SO. CONTINUE RINSING.

P321+P312 – SPECIFIC TREATMENT: CALL A POISON CENTER OR DOCTOR IF YOU FEEL UNWELL. REFER TO SECTION 4 OF THIS SDS.

P332+P337+P313 – IF SKIN IRRITATION OCCURES OR IF EYE IRRITATION PERSISTS, GET MEDICAL ATTENTION.

PRECAUTIONARY STATEMENTS CONTI.

P362 – TAKE OFF CONTAMINATED CLOTHING AND WASH BEFORE REUSE

PREVENTION:

P261 AVOID BREATHING DUST

P264 WASH HANDS AND OTHER EXPOSED

SKIN AREAS THOROUGHLY AFTER HANDLING

P271 USE ONLY OUTDOORS OR IN A WELL VENTILATED AREA

P280 WEAR PROTECTIVE GLOVES, PROTECTIVE CLOTHING AND EYE PROTECTION

STORAGE AND DISPOSAL STATEMENTS:

P405+P403+P233 – STORE LOCKED UP IN A WELL VENTILATED PLACE. KEEP CONTAINER TIGHTLY CLOSED

P501 – DISPOSE OF CONTENTS/CONTAINER IN ACCORDANCE WITH LOCAL/REGIONAL/ NATIONAL REGULATION.

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

COMPONENT	CAS NUMBER	% WEIGHT	EC NUMBER	INDEX NUMBER
SURFACTANTS	PROPRIETARY	≤ 8%		
2-BUTOXYETHANOL	111-76-2	≤ 5%	203-905-0	603-014-00-0
SODIUM METASILICATE	6834-92-0	≤ 5%	229-912-9	014-010-00-8
TETRASODIUM EDTA	64-02-8	≤ 2%	200-573-9	

COMPONENT	GHS CLASSIFICATION
SURFACTANTS	H315, H318, H401, H412
2-BUTOXYETHANOL	H227, H302, H312, H315, H319, H332
SODIUM METASILICATE	H290, H314, H335
TETRASODIUM EDTA	H302, H318

SECTION 4: FIRST AID MEASURES

DESCRIPTION OF FIRST AID MEASURES

INHALATION: IF PRODUCT MIST OR VAPOR CAUSES RESPIRATORY IRRITATION OR DISTRESS, MOVE THE EXPOSED PERSON TO FRESH AIR IMMEDIATELY. IF BREATHING IS DIFFICULT OR IRREGULAR, ADMINISTER OXYGEN; IF RESPIRATORY ARREST OCCURES, START ARTIFICIAL RESPIRATION BY TRAINED PERSONNEL. IF UNCONCIOUS, MAINTAIN AN OPEN AIRWAY. LOOSEN TIGHT CLOTHING SUCH AS A COLLAR, TIE, BELT OR WAISTBAND. IF SYMPTOMS PERSIST OR IF THE VICTIM FEELS UNWELL SEEK MEDICAL ATTENTION.

EYES: IMMEDIATELY FLUSH EYES WITH LARGE AMOUNTS OF WATER OR SALINE SOLUTION FOR AT LEAST 15 MINUTES, OCCASIONALLY LIFTING THE UPPER AND LOWER LIDS. REMOVE CONTACT LENSES IF PRESENT AND EASY TO DO, AFTER FIRST 2 MINUTES AND CONTINUE RINSING. SEEK IMMEDIATE MEDICAL ATTENTION, PREFERABLY FROM AN OPHTHAMOLOGIST.

SKIN: FLUSH SKIN WITH LARGE AMOUNTS OF WATER WHILE REMOVING CONTAMINATED CLOTHING. WASH THE AFFECTED AREA WITH SOAP AND WATER FOLLOWED BY THOROUGH RINSING. WASH CONTAMINATED CLOTHING AND SHOES BEFORE REUSE. IF IRRITATION PERSISTS OR IF THE VICTIM FEELS UNWELL, SEEK MEDICAL ATTENTION.

INGESTION: RINSE MOUTH WITH WATER IF THE VICTIM IS CONSCIOUS. REMOVE DENTURES IF PRESENT. GIVE 2 GLASSES OF WATER (MAXIMUM) IF THE VICTIM IS CONSCIOUS, ALERT AND ABLE TO SWALLOW.

DO NOT INDUCE VOMITING UNLESS DIRECTED TO DO SO BY MEDICAL PERSONNEL. VOMITING MAY OCCUR SPONTANEOUSLY. TO PREVENT ASPIRATION OF MATERIAL INTO THE LUNGS, LAY THE VICTIM ON ONE SIDE WITH THE HEAD LOWER THAN THE WAIST. NEVER GIVE ANYTHING BY MOUTH TO AN UNCONSCIOUS OR CONVULSING PERSON. DO NOT LEAVE VICTIM UNATTENDED. SEEK IMMEDIATE MEDICAL ATTENTION.

MOST IMPORTANT SYMPTOMS AND EFFECTS BOTH ACUTE AND DELAYED

POTENTIAL HEALTH SYMPTOMS AND EFFECTS

EYES: CAUSES SERIOUS EYE IRRITATION WITH INFLAMMATION, TEARING, SWELLING AND PAIN. PROLONGED CONTACT WITH EYE MAY CAUSE BURNS TO THE CONJUNCTIVA AND CORNEA. PROLONGED CONTACT MAY CAUSE EYE DAMAGE AND VISUAL IMPAIRMENT. VAPOR OF MIST CAN CAUSE EYE IRRITATION.

SKIN: CAUSES SKIN IRRITATION WITH LOCALIZED REDNESS, ITCHING, AND DISCOMFORT. PROLONGED CONTACT WITH UNPROTECTED SKIN MAY CAUSE BURNS, DEFATTING OF THE SKIN AND OR DERMATITIS. MAY BE HARMFUL IF ABSORBED THROUGH SKIN.

INHALATION: INHALATION OF MIST OR VAPOR MAY CAUSE RESPIRATORY IRRITATION. CAUSES HEADACHE, NASAL IRRITATION, COUGH AND BREATHING DIFFICULTY.

INGESTION: MAY BE HARMFUL IF SWALLOWED. CAUSES IRRITATION OF THE GASTROINTESTINAL TRACT WITH NAUSEA, ABDOMINAL PAIN, VOMITING AND DIARRHEA. MAY CAUSE BURNS TO THE ORAL CAVITY AND DIGESTIVE TRACT.

CHRONIC: INDIVIDUALS WITH PRE-EXISTING SKIN CONDITIONS AND RESPIRATORY DISORDERS MAY BE MORE SUSCEPTIBLE TO THE EFFECTS OF THIS PRODUCT. PROLONGED OR REPEATED SKIN CONTACT MAY CAUSE DRYING AND CRACKING OF THE SKIN, DERMATITIS OR AGGRAVATE EXISTING SKIN CONDITIONS. CHRONIC EYE CONTACT MAY CAUSE CONJUNCTIVITIS AND EYE DAMAGE. 2-BUTOXYETHANOL IS A KNOWN ANIMAL CARCINOGEN. REFER TO SECTION 11.2

INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT IF NEEDED

ADVICE TO DOCTOR AND HOSPITAL PERSONNEL

TREAT SYMPTOMATICALLY AND SUPPORTIVELY.

SECTION 5: FIRE FIGHTING MEASURES

SUITABLE EXTINGUISHING MEDIA

SUITABLE METHODS FOR EXTINCTION: USE EXTINGUISHING MEDIA SUTABLE FOR THE SURROUNDING FIRE

UNSUITABLE METHODS OF EXTENCTION: NO LIMATIIONS OF EXTINGUISHING AGENTS ARE GIVEN FOR THIS MATERIAL

SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE

CLOSED CONTAINERS MAY EXPLODE DUE TO A BUILD UP OF PRESSURE WHEN EXPOSED TO EXTREME HEAT. DURING EMERGENCY CONDITIONS, OVEREXPOSURE TO DECOMPOSITION PRODUCTS MAY CAUSE A HEALTH HAZARD. SYMPTOMS MAY NOT BE IMMEDIATELY APPARENT OR MAY BE DELAYED. OBTAIN MEDICAL ATTENTION.

EXPLOSION HAZARDS: THIS PRODUCT IS NOT AN EXPLOSION HAZARD.

ADVICE TO FIREFIGHTERS

FULL PROTECTIVE EQUIPMENT INCLUDING SELF-CONTAINED BREATHING APPARATUS SHOULD BE USED.

WATER MAY BE USED TO COOL CLOSED CONTAINERS TO PREVENT PRESSURE BUILDUP AND POSSIBLE AUTO-IGNITION OR EXPLOSION WHEN EXPOSED TO EXTREME HEAT. IF POSSIBLE, WATER CONTAMINATED BY THIS MATERIAL SHOULD BE CONTAINEDFROM BEING DISCHARGED TO ANY WATERWAY, SEWER, OR DRAIN TO PREVENT ENVIORNMENTAL CONTAMINATION.

SECTION 6: ACCIDENT RELEASE MEASURES

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

EVACUATE NON-ESSENTIAL PERSONNEL. WEAR APPROPRIATE PROTECTIVE CLOTHING AND EQUIPMENT DESIGNATED IN SECTION 8.2 VENTILATE THE AREA. REMOVE ALL SOURCES OF IGNITION. NO SMOKING. CLEAN UP SPILLS IMMEDIATELY. SPILL CREATES A SLIP HAZARD.



ENVIRONMENTAL PRECAUTIONS

AVOID DISPERSAL OF SPILLED MATERIAL OR RUNOFF AND PREVENT CONTACT WITH SOIL

AND ENTRY INTO DRAINS, SEWERS OR WATERWAYS.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING UP
 APPROACH SPILL FROM AN UPWIND DIRECTION. DO NOT FLUSH SPILLS DOWN THE DRAIN. COVER DRAINS AND CONTAIN SPILL. COVER SPILL WITH A LARGE QUANTITY OF INERT ABSORBENT. DO NOT USE COMBUSTIBLE MATERIAL SUCH AS SAWDUST. COLLECT MATERIAL AND PLACE INTO AN APPROVED CONTAINER FOR PROPER DISPOSAL. OBSERVE POSSIBLE MATERIAL RESTRICTIONS SECTION 7.2 & 10.5. DO NOT ALLOW MATERIAL OR RUNOFF FROM RINSING CONTAMINATED AREAS TO ENTER FLOOR DRAINS OR STORM DRAINS AND DITCHES THAT LEAD TO WATERWAYS. DISPOSE OF CONTENTS AND CONTAINERS VIA A LICENSED WASTE DISPOSAL CONTRACTOR. FOR INDICATIONS ABOUT WASTE TREATMENT, SEE SECTION 13.

SECTION 7: HANDLING AND STORAGE

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES: EVACUATE NON-ESSENTIAL PERSONNEL. WEAR APPROPRIATE PROTECTIVE CLOTHING AND EQUIPMENT DESIGNATED IN SECTION 8.2. VENTILATE THE AREA. REMOVE ALL SOURCES OF IGNITION. NO SMOKING. CLEAN UP SPILLS IMMEDIATELY. SPILL CREATES A SLIP HAZARD.

PRECAUTIONS FOR SAFE HANDLING: WEAR ALL APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT SPECIFIED IN SECTION 8.2. DO NOT GET IN EYES OR ON SKIN OR CLOTHING. DO NOT INHALE MIST OR VAPOR. NO SMOKING. IF NORMAL USE OF MATERIAL PRESENTS A RESPIRATORY HAZARD, USE ONLY ADEQUATE VENTILATION OR WEAR AN APPROPRIATE RESPIRATOR. WASH CONTAMINATED CLOTHING AND SHOES THOROUGHLY BEFORE REUSE.

ADVICE ON PROTECTION AGAINST FIRE AND EXPLOSION: THIS PRODUCT IS NOT A FIRE OR EXPLOSION HAZARD.

CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES: STORE IN DRY, COOL, WELL-VENTILATED AREAS AWAY FROM INCOMPATIBLE MATERIALS (SEE SECTION 10.5), FOOD, AND DRINK. KEEP FROM FREEZING. TRANSFER ONLY TO APPROVED CONTAINERS HAVING CORRECT LABELING. KEEP CONTAINERS TIGHTLY CLOSED WHEN NOT IN USE. PROTECT CONTAINERS AGAINST PHYSICAL DAMAGE. CONTAINERS THAT HAVE BEEN OPENED MUST BE CAREFULLY RESEALED AND KEPT UPRIGHT TO PREVENT LEAKAGE. CONTAINERS ARE HAZARDOUS WHEN EMPTY AS THEY CONTAIN PRODUCT RESIDUE. USE APPROPRIATE CONTAINMENT TO AVOID ENVIRONMENTAL CONTAMINATION. VENTILATE CLOSED AREAS. KEEP LOCKED UP AND OUT OF REACH OF CHILDREN.

SPECIFIC END USES: APART FROM THE USES MENTIONED IN SECTION 1.2, NO OTHER SPECIFIC USES ARE STIPULATED

SECTION 8: EXPOSURE CONTROL/PERSONAL PROTECTION

CONTROL PARAMETERS
 OCCUPATIONAL EXPOSURE LIMIT VALUES

CAS NUMBER	INGREDIENT	OSHA PEL	ACGIH TLV	NIOSH
111-76-2	2-Butoxyethanol	50 ppm; 240 mg/m ³ TWA	20 ppm; 240 mg/m ³ TWA; skin	50 ppm; 24 mg/m ³ TWA 700 ppm IDLH; skin

A "skin" notation following the inhalation exposure guideline refers to the potential for dermal absorption of the material, including eyes and mucous membranes, either by direct contact with vapors or by direct skin contact. It is intended to alert the reader that inhalation may not be the only route of exposure and that measures to minimize dermal exposure should be considered.

EXPOSURE CONTROLS

ENGINEERING MEASURES: TECHNICAL MEASURES AND APPROPRIATE WORKING OPERATIONS SHOULD BE GIVEN PRIORITY OVER THE USE OF PERSONAL PROTECTIVE EQUIPMENT. USE ADEQUATE VENTILATION. LOCAL EXHAUST IS PREFERABLE. REFER TO SECTION 7.1.

INDIVIDUAL PROTECTION MEASURES: WEAR PROTECTIVE CLOTHING TO PREVENT REPEATED OR PROLONGED CONTACT WITH PRODUCTS. PROTECTIVE CLOTHING NEEDS TO BE SELECTED SPECIFICALLY FOR THE WORK-PLACE, DEPENDING ON CONCENTRATIONS AND QUANTITIES OF HAZARDOUS SUBSTANCES HANDLED. THE CHEMICAL RESISTANCE OF THE PROTECTIVE EQUIPMENT SHOULD BE ENQUIRED AT THE REPRESENTATIVE SUPPLIER.

HYGIENE MEASURES: FACILITIES STORING OR USING THIS MATERIAL SHOULD BE EQUIPPED WITH AN EYE-WASH STATION AND SAFETY SHOWER. CHANGE CONTAMINATED CLOTHING. PREVENTIVE SKIN PROTECTION IS RECOMMENDED. WASH HANDS THOROUGHLY AFTER USE, BEFORE EATING, DRINKING, SMOKING, OR USING THE LAVATORY.

EYE/FACE PROTECTION: WEAR SAFETY GLASSES WITH UNPERFORATED SIDE SHIELDS OR PROTECTIVE SPLASH GOGGLES DURING USE.

HAND PROTECTION: WEAR GLOVES RECOMMENDED BY GLOVE SUPPLIER FOR PROTECTION AGAINST MATERIALS IN SECTION 3. GLOVES SHOULD BE IMPERMEABLE TO CHEMICALS AND OIL. BREAKTHROUGH TIME OF SELECTED GLOVES MUST BE GREATER THAN THE INTENDED USE PERIOD.

SKIN PROTECTION: WEAR PROTECTIVE CLOTHING. WEAR PROTECTIVE BOOTS IF THE SITUATION REQUIRES.

RESPIRATORY PROTECTION: ALWAYS USE AN APPROVED RESPIRATOR WHEN VAPOR/AEROSOLS EXCEED PERMISSIBLE EXPOSURE LIMITS. WHERE RISK ASSESSMENT SHOWS AIR-PURIFYING RESPIRATORS ARE APPROPRIATE, USE A HALF-MASK RESPIRATOR WITH MULTI-PURPOSE COMBINATION (US) OR TYPE ABEK (EN 14387) RESPIRATOR CARTRIDGES AS A BACKUP TO ENGINEERING CONTROLS. IF THE RESPIRATOR IS THE SOLE MEANS OF PROTECTION, USE A FULL-FACE SUPPLIED AIR RESPIRATOR. USE RESPIRATORS AND COMPONENTS TESTED AND APPROVED UNDER APPROPRIATE GOVERNMENT STANDARDS SUCH AS NIOSH (US) OR CEN (EU). FOLLOW OSHA RESPIRATOR REGULATIONS FOUND IN 29 CFR 1910.134 OR EUROPEAN STANDARD EN 149

ENVIRONMENTAL EXPOSURE CONTROLS: DO NOT EMPTY INTO DRAINS.

PPE must not be considered a long-term solution to exposure control. PPE usage must be accompanied by employer programs to properly select, maintain, clean, fit, and use. Consult a competent industrial hygiene resource to determine hazard potential and/or the PPE manufacturers to ensure adequate protection



SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

INFORMATION ON BASIC PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE:	CLEAR, GREEN LIQUID
ODOR	CHARACTERISTIC
ODOR THRESHOLD	NO DATA AVAILABLE
MOLECULAR WEIGHT	NO DATA AVAILABLE
CHEMICAL FORMULA	NO DATA AVAILABLE
PH	10.0-11.0
FREEZING/MELTING POINT	NO DATA AVAILABLE
INITIAL BOILING POINT	100 °C(212 °F)
EVAPORATION RATE	NO DATA AVAILABLE
FLAMMABILITY (SOLID, GAS)	NOT APPLICABLE
FLASH POINT	NOT APPLICABLE
AUTOIGNITION TEMPERATURE	NO DATA AVAILABLE
DECOMPOSITION TEMPERATURE	NO DATA AVAILABLE
LOWER EXPLOSIVE LIMIT	(LEL) NO DATA AVAILABLE
UPPER EXPLOSIVE LIMIT	(UEL) NO DATA AVAILABLE
VAPOR PRESSURE	NO DATA AVAILABLE
VAPOR DENSITY	NO DATA AVAILABLE
DENSITY	1.013-1.025G/ML (8.45-9.55LB/GAL)
VISCOSITY	NO DATA AVAILABLE
SOLUBILITY IN WATER	COMPLETELY MISCIBLE
PARTITION COEFFICIENT (N-OCTANOL/WATER)	NO DATA AVAILABLE
OXIDIZING PROPERTIES	NOT APPLICABLE
EXPLOSIVE PROPERTIES	NOT APPLICABLE
VOLATILES BY WEIGHT @ 21 °C	98.71%
VOC(WT.%)	0.507(5.07G/ML;0.042LB/GAL)

SECTION 10: STABILITY AND REACTIVITY

REACTIVITY

NO SPECIAL REACTIVITY HAS BEEN REPORTED DURING NORMAL CONDITIONS OF HANDLING AND USE.

CHEMICAL STABILITY

THIS MATERIAL IS STABLE UNDER RECOMMENDED STORAGE AND HANDLING CONDITIONS.

POSSIBILITY OF HAZARDOUS REACTIONS

HAZARDOUS POLYMERIZATION WILL NOT OCCUR.

CONDITIONS TO AVOID

AVOID TEMPERATURE EXTREMES AND CONTACT WITH INCOMPATIBLE MATERIALS.

INCOMPATIBLE MATERIALS

STRONG OXIDIZING AGENTS, ACIDS

HAZARDOUS DECOMPOSITION PRODUCTS

THERMAL DECOMPOSITION PRODUCTS MAY INCLUDE OXIDES OF CARBON, AND SODIUM OXIDE FUME

SECTION II: TOXICOLOGICAL INFORMATION

11.1 INFORMATION ON TOXICOLOGICAL EFFECTS

ACUTE ORAL TOXICITY	LD50, RAT: > 5,000 MG/KG [CALCULATED]
ACUTE INHALATION TOXICITY	NO DATA AVAILABLE
ACUTE DERMAL TOXICITY	LD50, RAT: > 5,000 MG/KG [CALCULATED]
SKIN IRRITATION	CAUSES SKIN IRRITATION.
EYE IRRITATION	CAUSES SERIOUS EYE IRRITATION
SENSITIZATION	NO DATA AVAILABLE
CARCINOGENICITY	NO DATA AVAILABLE
GERM CELL MUTAGENICITY	NO DATA AVAILABLE
REPRODUCTIVE TOXICITY	NO DATA AVAILABLE
SPECIFIC ORGAN TOXICITY-SINGLE EXPOSURE	MAY CAUSE RESPIRATORY IRRITATION.
SPECIFIC ORGAN TOXICITY-REPEATED EXPOSURE	NO DATA AVAILABLE
ASPIRATION HAZARD	NO DATA AVAILABLE

11.2 FURTHER INFORMATION

2-BUTOXYETHANOL(CAS #111-76-2): IARC GROUP 3 CARCINOGEN-NOT CLASSIFIABLE AS TO ITS CARCINOGENICITY TO HUMANS. NOT LISTED AS A CARCINOGEN BY ACGIH, NTP OR OSHA. IN LONG-TERM ANIMAL STUDIES WITH 2-BUTOXYETHANOL, SMALL BUT STATISTICALLY SIGNIFICANT INCREASES IN TUMORS WERE OBSERVED IN MICE BUT NOT RATS. THE EFFECTS ARE NOT BELIEVED TO BE RELEVANT TO HUMANS.

2-BUTOXYETHANOL INHALATION EXPOSURE IN LABORATORY ANIMALS HAS BEEN FOUND TO REDUCE BODY WEIGHT GAIN AND FOOD CONSUMPTION IN ADDITION TO HEMOLYSIS. AFTER EXPOSURE WAS DISCONTINUED, THESE EFFECTS IN ANIMALS DISAPPEARED. ADVERSE REPRODUCTIVE OR BIRTH EFFECTS WERE NOT REPORTED IN ANIMALS EXCEPT WHEN EXPOSURES WERE HIGH ENOUGH TO CAUSE SIGNIFICANT MATERNAL TOXICITY. IN ANIMALS, HEMOLYSIS (RED BLOOD CELL BREAKAGE) AND SECONDARY EFFECTS TO THE KIDNEYS AND LIVER HAVE BEEN REPORTED. HUMAN RED BLOOD CELLS HAVE BEEN SHOWN TO BE SIGNIFICANTLY LESS SENSITIVE TO HEMOLYSIS THAN THOSE OF RODENTS AND RABBITS.

NO DATA IS AVAILABLE REGARDING THE MUTAGENICITY OR TERATOGENICITY OF THIS PRODUCT, NOR IS THERE ANY AVAILABLE DATA THAT INDICATES THAT IT CAUSES ADVERSE DEVELOPMENTAL OR FERTILITY EFFECTS.

HANDLE IN ACCORDANCE WITH GOOD INDUSTRIAL HYGIENE AND SAFETY PRACTICE

SECTION 12: ECOLOGICAL INFORMATION

TOXICITY

LARGE SPILLS OR DISCHARGES OF THIS PRODUCT MAY BE HARMFUL TO AQUATIC LIFE. LARGE SPILLS OR DISCHARGES MAY INCREASE THE PH OF AQUATIC SYSTEMS AND CAN BE HARMFUL TO AQUATIC LIFE AND SOIL MICROORGANISMS. 2-BUTOXYETHANOL IS HARMFUL TO ALGAE AND HIGHER AQUATIC PLANTS.

PERSISTENCE AND DEGRADABILITY

ORGANIC COMPONENTS IN THIS PRODUCT ARE BIODEGRADABLE. INORGANIC SUBSTANCES ARE NOT BIODEGRADABLE. METHODS FOR THE DETERMINATION OF BIODEGRADABILITY ARE NOT APPLICABLE TO INORGANIC SUBSTANCES.

BIOACCUMULATION POTENTIAL
MOBILITY IN SOIL

NO DATA AVAILABLE
NO DATA AVAILABLE

RESULTS OF PBT AND VPVB ASSESSMENT

THIS MATERIAL DOES NOT CONTAIN SUBSTANCES THAT ARE PERSISTENT, BIOACCUMULATIVE, AND TOXIC (PBT) OR VERY PERSISTENT AND VERY BIOACCUMULATIVE (VPVB).

ENDOCRINE DISRUPTING PROPERTIES

THIS MATERIAL DOES NOT CONTAIN COMPONENTS CONSIDERED TO HAVE ENDOCRINE DISRUPTING PROPERTIES ACCORDING TO REACH ARTICLE 57(F) OR COMMISSION DELEGATED REGULATION (EU) 2017/2100 COMMISSION REGULATION (EU) 2018/605 AT LEVELS OF 0.1% OR HIGHER.

OTHER EFFECTS

ADDITIONAL ECOLOGICAL INFORMATION DO NOT ALLOW MATERIAL TO RUN INTO SURFACE WATERS, WASTE WATER, OR SOIL. AN ENVIRONMENTAL HAZARD CANNOT BE EXCLUDED IN THE EVENT OF UNPROFESSIONAL HANDLING OR DISPOSAL

SECTION 13: DISPOSAL CONSIDERATIONS

WASTE TREATMENT METHODS

METHODS OF DISPOSAL: THE GENERATION OF WASTE SHOULD BE AVOIDED OR MINIMIZED WHENEVER POSSIBLE. EMPTY CONTAINERS OR LINERS MAY RETAIN SOME PRODUCT RESIDUES. THIS MATERIAL AND ITS CONTAINER MUST BE DISPOSED OF IN A SAFE WAY. DISPOSE OF SURPLUS AND NON-RECYCLABLE PRODUCTS IN ACCORDANCE WITH NATIONAL, STATE AND LOCAL REGULATIONS. DISPOSAL OF THIS PRODUCT, SOLUTIONS, AND ANY BY-PRODUCTS SHOULD AT ALL TIMES COMPLY WITH THE REQUIREMENTS OF ENVIRONMENTAL PROTECTION AND WASTE DISPOSAL LEGISLATION AND ANY REGIONAL LOCAL AUTHORITY REQUIREMENTS. AVOID DISPERSAL OF SPILT MATERIAL AND RUNOFF AND CONTACT WITH SOIL, WATERWAYS, DRAINS, AND SEWERS.

RCRA F-SERIES: NO LISTINGS ABOVE THE REPORTABLE THRESHOLD (DE MINIMIS)

RCRA U-SERIES: NO LISTINGS ABOVE THE REPORTABLE THRESHOLD (DE MINIMIS)

SECTION 14: TRANSPORT INFORMATION

NOTE: TRANSPORTATION INFORMATION PROVIDED IS FOR REFERENCE ONLY. CUSTOMERS ARE URGED TO CONSULT 49 CFR 100-177, IMDG, IATA, EC, UNITED NATIONS TDG, AND WHMIS (CANADA) TDG INFORMATION MANUALS FOR DETAILED REGULATIONS AND EXCEPTIONS COVERING SPECIFIC CONTAINER SIZES, PACKAGING MATERIALS, AND METHODS OF SHIPPING.

USA DOT (GROUND TRANSPORTATION)
IMO/IMDG (WATER TRANSPORTATION)
ICAO/IATA (AIR TRANSPORTATION)
RID/ADR (RAIL TRANSPORTATION)

NOT REGULATED FOR TRANSPORT
NOT REGULATED FOR TRANSPORT
NOT REGULATED FOR TRANSPORT
NOT REGULATED FOR TRANSPORT

SECTION 15: REGULATORY INFORMATION**SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS/LEGISLATION SPECIFIC FOR SUBSTANCE OR MIXTURE****U.S. FEDERAL REGULATIONS**

OSHA HAZARD COMMUNICATION STANDARD: THIS MATERIAL IS CLASSIFIED AS HAZARDOUS IN ACCORDANCE WITH OSHA 29 CFR 1910.1200.

TOXIC SUBSTANCE CONTROL ACT (TSCA) INVENTORY: ALL SUBSTANCES IN THIS PRODUCT ARE LISTED ON THE TSCA INVENTORY. THIS PRODUCT IS NOT SUBJECT TO TSCA 12(B) EXPORT NOTIFICATION.

OSHA PROCESS SAFETY MANAGEMENT STANDARD: THIS PRODUCT IS NOT REGULATED UNDER OSHA PSM STANDARD 29 CFR 1910.119.

EPA RISK MANAGEMENT PLANNING STANDARD: THIS PRODUCT IS NOT REGULATED UNDER EPA RMP STANDARD (RMP) 40 CFR PART 68.

EPA FEDERAL INSECTICIDE, FUNGICIDE AND RODENTICIDE ACT: THIS PRODUCT IS NOT A REGISTERED PESTICIDE UNDER THE FIFRA, 40 CFR PART 150.

TOXIC SUBSTANCE CONTROL ACT (TSCA) INVENTORY: ALL SUBSTANCES IN THIS PRODUCT ARE LISTED ON THE TSCA INVENTORY. THIS PRODUCT IS NOT SUBJECT TO TSCA 12(B) EXPORT NOTIFICATION.

DRUG ENFORCEMENT ADMINISTRATION (DEA) LIST 2, ESSENTIAL CHEMICALS (21 CFR 1310.02(B)) AND 1310.4(F)(2)) AND CHEMICAL CODE NUMBER: NO LISTINGS.

DRUG ENFORCEMENT ADMINISTRATION (DEA) LISTS 1 & 2, EXEMPT CHEMICAL MIXTURES (21 CFR 1310.12(C)) AND CODE NUMBER: NO LISTINGS.

DEPARTMENT OF HOMELAND SECURITY (DHS) CHEMICAL FACILITY ANTI-TERRORISM STANDARDS (CFATS) CHEMICALS: NO LISTINGS.

SUPERFUND AMENDMENTS AND REAUTHORIZATION ACT (SARA)

SARA SECTION 311/312 HAZARD CATEGORIES: CAUSES SKIN IRRITATION AND SERIOUS EYE IRRITATION. MAY CAUSE RESPIRATORY IRRITATION.

SARA 313 INFORMATION: GLYCOL ETHERS (SARA CODE N230) ARE SUBJECT TO THE REPORTING LEVELS ESTABLISHED BY SECTION 313 OF THE EMERGENCY PLANNING AND COMMUNITY RIGHT-TO-KNOW ACT OF 1986.

SARA 302/304 EXTREMELY HAZARDOUS SUBSTANCE: NONE OF THE COMPONENTS OF THIS PRODUCT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED BY THESE SECTIONS OF TITLE III OF SARA.

SARA 302/304 EMERGENCY PLANNING & NOTIFICATION: NONE OF THE COMPONENTS OF THIS PRODUCT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED BY THESE SECTIONS OF TITLE III OF SARA.

COMPREHENSIVE RESPONSE COMPENSATION AND LIABILITY ACT (CERCLA): THIS PRODUCT CONTAINS THE FOLLOWING CERCLA REPORTABLE SUBSTANCES: GLYCOL ETHERS – THERE IS NO RQ ASSIGNED TO THIS BROAD CLASS, ALTHOUGH THE CLASS IS A CERCLA HAZARDOUS SUBSTANCES. REFER TO 50 FEDERAL REGISTER 13456 (APRIL 4, 1985).

CLEAN AIR ACT (CAA)

THIS PRODUCT DOES NOT CONTAIN HAZARDOUS AIR POLLUTANTS (HAPS) DESIGNATED IN CAA SECTION 112 (B).

THIS PRODUCT DOES NOT CONTAIN CLASS I OZONE DEPLETERS.

THIS PRODUCT DOES NOT CONTAIN CLASS II OZONE DEPLETERS.

CLEAN WATER ACT (CWA)

GLYCOL ETHERS (EDF-109) AND SODIUM HYDROXIDE ARE HAZARDOUS SUBSTANCES.

THIS PRODUCT DOES NOT CONTAIN ANY PRIORITY POLLUTANTS.

THIS PRODUCT DOES NOT CONTAIN ANY TOXIC POLLUTANTS.

U.S. STATE REGULATIONS**CALIFORNIA PROP 65, SAFE DRINKING WATER AND TOXIC ENFORCEMENT ACT OF 1986**

THIS PRODUCT CONTAINS NO CHEMICAL(S) KNOWN TO THE STATE OF CALIFORNIA TO CAUSE CANCER, BIRTH DEFECTS, OR REPRODUCTIVE HARM IN CONCENTRATIONS THAT EXCEED THE THRESHOLD (DE MINIMIS) REPORTING LEVELS ESTABLISHED UNDER PROPOSITION 65.

OTHER U.S. STATE INVENTORIES

2-BUTOXYETHANOL (CAS #111-76-2) IS LISTED ON THE FOLLOWING STATE HAZARDOUS SUBSTANCE INVENTORIES, RIGHT-TO-KNOW LISTS, AND/OR AIR QUALITY/AIR POLLUTANTS LISTS: CA, MN, PA, RI, WI.

CANADA

WHMIS HAZARD CLASSIFICATION: NO DATA AVAILABLE

CANADIAN NATIONAL POLLUTANT RELEASE INVENTORY (NPRI): 2-BUTOXYETHANOL IS LISTED ON THE NPRI.

EUROPEAN ECONOMIC COMMUNITY

WGK, GERMANY (WATER DANGER/PROTECTION): 1 (SLIGHTLY HAZARDOUS TO WATER)

Global Chemical Inventory Lists

COUNTRY	INVENTORY NAME	LISTED
Canada	Domestic Substance List (DSL)	Yes
Canada	Non-Domestic Substance List (NDSL)	No
Europe	Inventory of New and Existing Chemicals (EINECS)	Yes
United States	Toxic Substance Control Act (TSCA)	Yes
Australia	Australian Inventory of Chemical Substances (AICS)	Yes
New Zealand	New Zealand Inventory of Chemicals (NZIoC)	Yes
China	Inventory of Existing Chemical Substances in China (IECSC)	Yes
Japan	Inventory of Existing and New Chemical Substances (ENCS)	Yes
Korea	Existing Chemicals List (ECL)	Yes
Philippines	Philippines Inventory of Chemicals and Chemical Substances (PICCS)	Yes

YES - ALL COMPONENTS OF THIS PRODUCT COMPLY WITH THE INVENTORY REQUIREMENTS ADMINISTERED BY THE GOVERNING COUNTRY.

NO - ONE OR MORE COMPONENTS OF THIS PRODUCT ARE NOT ON THE INVENTORY OR ARE EXEMPT FROM LISTING.

CHEMICAL SAFETY ASSESSMENT: A CHEMICAL SAFETY ASSESSMENT WAS NOT CARRIED OUT FOR THIS PRODUCT.

SECTION 16: OTHER INFORMATION**HMIS HAZARDOUS CLASSIFICATION**

HEALTH HAZARD: 2

FLAMMABILITY: 0

INSTABILITY: 0

PERSONAL PROTECTION: C

DISCLAIMER OF RESPONSIBILITY

THE INFORMATION ON THIS SDS WAS OBTAINED FROM SOURCES WHICH WE BELIEVE ARE RELIABLE. HOWEVER, THE INFORMATION IS PROVIDED WITHOUT ANY WARRANTY, EXPRESSED OR IMPLIED, REGARDING ITS CORRECTNESS. SOME INFORMATION PRESENTED, AND CONCLUSIONS DRAWN HEREIN ARE FROM SOURCES OTHER THAN DIRECT TEST DATA ON THE SUBSTANCE ITSELF. THE CONDITIONS OR METHODS OF HANDLING, STORAGE, USE AND DISPOSAL OF THE PRODUCT ARE BEYOND OUR CONTROL AND MAY BE BEYOND OUR KNOWLEDGE. FOR THIS AND OTHER REASONS, WE DO NOT ASSUME DAMAGE OR EXPENSE ARISING OUT OF OR IN ANY WAY RESPONSIBILITY AND EXPRESSLY DISCLAIM LIABILITY FOR LOSS, CONNECTED WITH HANDLING, STORAGE, USE, OR DISPOSAL OF THIS PRODUCT. IF THE PRODUCT IS USED AS A COMPONENT IN ANOTHER PRODUCT, THIS SDS INFORMATION MAY NOT BE APPLICABLE.

PREPARATION DATE: 25 JANUARY 2024, VERSION 1

Full Text of GHS Hazard Phrases Referenced in Section 3 (not covered in Section 2)

H227 – Combustible liquid
 H302 – Harmful if swallowed
 H312 – Harmful if inhaled
 H314 – Causes skin burns and serious eye damage
 H318 – Causes serious eye damage
 H332 – Harmful if inhaled
 H401 – Toxic to aquatic life
 H412 – Harmful to aquatic life with long lasting effects

Abbreviation Key

Abbreviation Full Form

ACGIH American Conference of Governmental Industrial Hygienists

ADR Accord Dangereux Routier (European regulations concerning the international transport of dangerous goods by road)

CAS Chemical Abstract Services

CFR Code of Federal Regulations

COC Cleveland Open Cup

DOT Department of Transportation

EC50 Half maximal effective concentration

EMS Emergency Response Procedures for Ships Carrying Hazardous Goods

EPA Environmental Protection Agency

ErC50 Reduction of Growth Rate

ERG Emergency Response Guide Book

FDA Food and Drug Administration

GHS Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

HCS Hazard Communication Standard

IARC International Agency for Research on Cancer

LDLo Lowest Lethal Dose

mppcf Millions of Particles Per Cubic Foot

NA North America

NAERG North American Emergency Response Guide Book

NIOSH National Institute for Occupational Safety & Health

NTP National Toxicology Program

OSHA Occupational Safety and Health Administration

PBT Persistent, Bioaccumulating and Toxic

PEL Permissible exposure limit

PMCC Pensky–Martens Closed Cup

ppm Parts Per Million

RCRA Resource Conservation and Recovery Act

RID Dangerous Goods by Rail

RQ Reportable Quantity

TCC/Tag Tagliabue Closed Cup

IATA International Air Transport Association

IC50 Half Maximal Inhibitory Concentration

ICAO International Civil Aviation Organization

IDLH Immediately Dangerous to Life and Health

IMDG International Maritime Dangerous Goods

IMO International Maritime Organization

LC50 50% Lethal Concentration

LD50 50% Lethal Dose

TLV Threshold Limit Value

TSCA Toxic Substance Control Act

TWA Time-weighted Average

UN United Nations

VOC Volatile Organic Compounds

vPvB Very Persistent and Very Bioaccumulating

WHMIS Workplace Hazardous Materials Information System